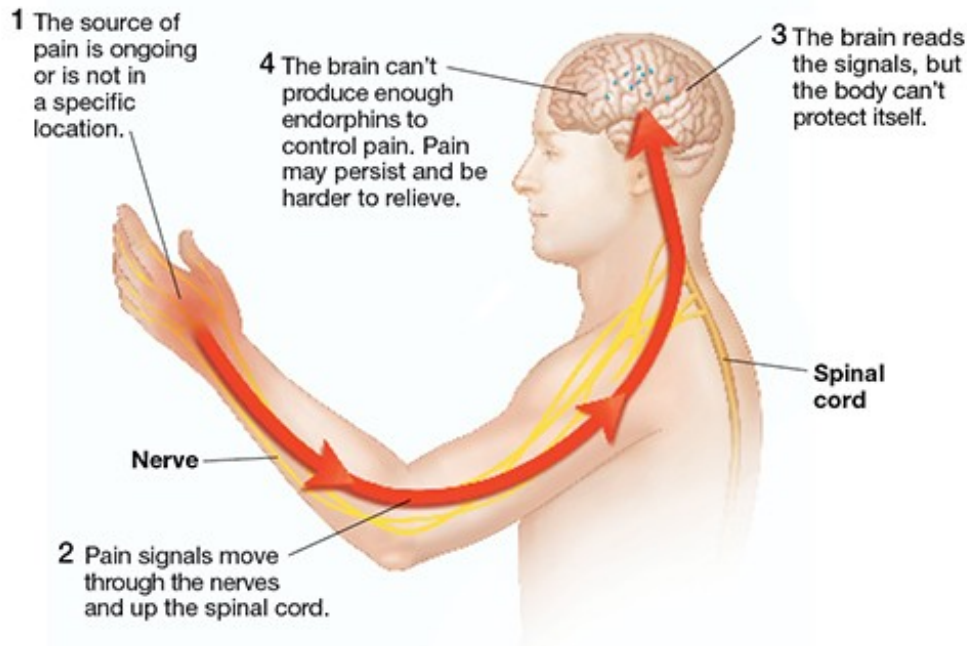


PAIN MANAGEMENT



Fact about pain

- Thinking positively can be an important part of managing chronic pain
- Researchers found that what people think can affect their pain
- People who expect less pain may feel less pain
- Intense feelings of love can also provide pain relief similar to the effects of painkillers, according to a Stanford study

Introduction

3

Definition of pain

- “An unpleasant sensory and emotional experience associated with actual or potential tissue damage or described in terms of such damage”
- Pain can affect all areas of a person’s life including sleep, thought, emotion, social, and activities of daily living
- Pain is **subjective experience**
 - Mostly described as ‘**whatever the patients says**’

Pathophysiology

4

Nociceptive Pain

- Usually considered as **acute pain**, is either:
- **Somatic:** localize directly to the site of injury (arising from skin, bone, joint, muscle, or connective tissue) or
 - ▣ Pts often describes it as: **dull**, sometimes sharp, **but very well localized**
- **Visceral:** diffuse, poorly differentiated, referred pain (arising from internal organs such as the large intestine or pancreas).
 - Feels like generalized aching or pressure, and is often harder to localize
 - Pt described it as**pressure, deep, squeezing, not well-localized**

Mechanism of pain

5

- Nociception can be expressed in terms of **stimulation transduction, transmission, perception, and modulation.**
- Stimulation of nociceptors (sensory nerve cells) is the first step leading to sensation of pain.
- These receptors are found in both somatic and visceral structures and are activated by **mechanical, thermal, and chemical** factors

Cont....

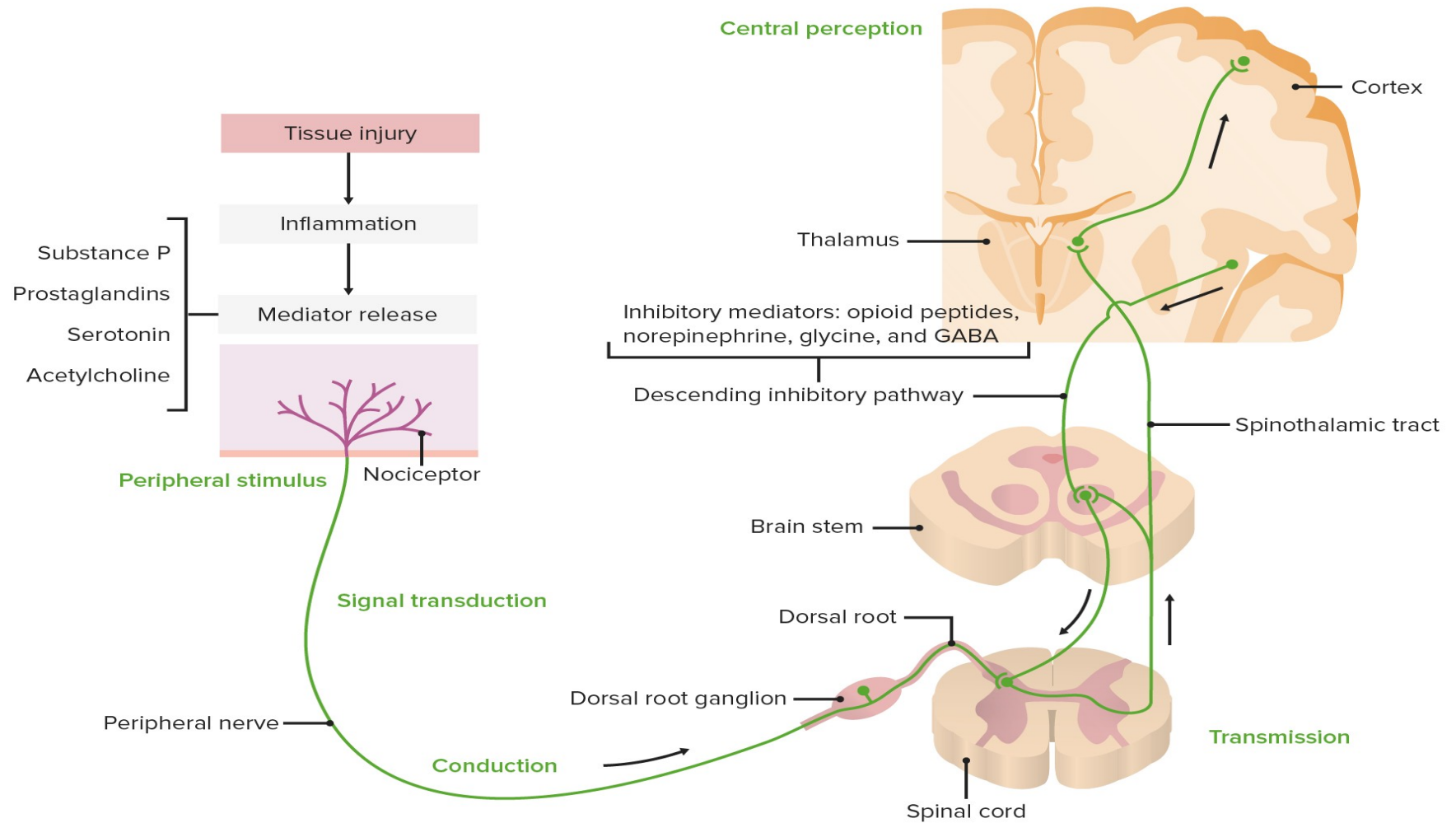
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- After receptor activation, impulses transmitted along afferent nerve fibers to the **spinal cord**
- The **thalamus** acts as a relay station/amplifies/ and passes the impulses to central structures where pain can be processed further.
- At this point in transmission, pain will be perceived (**pain perception**)

Cont.....

7

- The body modulates/inhibit/ pain through several processes:
 - ▣ The **endogenous opiate system** consists of NTs and receptors that are found throughout the CNS.
 - ▣ Endogenous opioids bind to opioid receptors and modulate the transmission of pain impulses.
 - ▣ The CNS also contains a **descending system** for control of pain transmission.
 - This system originates in the brain and can **inhibit synaptic pain transmission (GABA, NE, or serotonin)**



Neuropathic Pain

9

- **Often described in terms of chronic pain**
 - **Neuropathic pain** is a result of *nerve damage*, (e.g., postherpetic neuralgia, diabetic neuropathy, trigeminal neuralgia)
 - **Functional pain** refers to abnormal operation of the nervous system (e.g., fibromyalgia, irritable bowel syndrome, *tension-type headache*)

Classification of pain

10

There are numerous ways of classifying pain:

- **Type of pain** (nociceptive, neuropathic, inflammatory)
- **Pain intensity** (mild, moderate and severe)
- **Duration of pain** (acute and chronic pain)

Cont....

11

Acute pain

- Pain immediately following an injury to body
- Serving its adaptive purpose by warning individuals of disease states and potentially harmful situations
- Usually is nociceptive, although it can be neuropathic in nature.
- Serve a useful purpose in helping identifying the sites of injury.
- Usually self-limiting, subsides when the injury heals
- **Common causes of acute pain:** surgery, acute illness, trauma, labor, and medical procedures
- Lasting less than 3 to 6 months

Cont...

12

Chronic pain

- When pain fails to subside and **persist for months to years**- it leads to chronic pain state
- The pain can be nociceptive, neuropathic or both

Four subtypes

1. Pain that persists beyond normal healing time for acute injury
2. Pain associated with chronic diseases (osteoarthritis)
3. Pain without identifiable cause (fibromyalgia)
4. Cancer related pain

Characteristic	Acute Pain	Chronic Pain
Relief of pain	Highly desirable	Highly desirable
Dependence and tolerance to medication	Unusual	Common
Psychological component	Usually not present	Often a major problem
Organic cause	Common	Often not present
Environmental/family issues	Small	Significant
Insomnia	Unusual	Common component
Treatment goal	Cure	Functionality
Depression	Uncommon	Common

CLINICAL PRESENTATION OF PAIN

Acute

General

- Often obvious distress (e.g., trauma)

Symptoms

- Can be described as sharp, dull, shock-like, tingling, shooting, radiating, fluctuating in intensity, and varying in location (these occur in a timely relationship with an obvious noxious stimuli)

Signs

- Hypertension, tachycardia, diaphoresis, mydriasis, and pallor, but these signs are *not diagnostic*
- In some cases there are no obvious signs
- Comorbid conditions usually not present
- Outcome of treatment generally predictable

Laboratory Tests

- Pain is always subjective
- There are *no* specific laboratory tests for pain
- Pain is best diagnosed based on patient description and history

Chronic

General

- Can appear to have no noticeable suffering

Symptoms

- Can be described as sharp, dull, shock-like, tingling, shooting, radiating, fluctuating in intensity, and varying in location (these often occur *without* a timely relationship with an obvious noxious stimuli)
- Over time, the pain stimulus may cause symptoms that completely change (e.g., sharp to dull, obvious to vague)

Signs

- Hypertension, tachycardia, diaphoresis, mydriasis, and pallor are seldom present
- In most cases there are **NO** obvious signs
- Comorbid conditions often present (e.g., sleep problems, depression, relationship problems)
- Outcome of treatment often unpredictable

Laboratory Tests

- Pain is always subjective
- Pain is best diagnosed based on patient description and history
- There are *no* specific laboratory tests for pain; however, history and/or diagnostic proof of past trauma (e.g., computed tomography) or present disease state (e.g., autoantibodies) may be helpful in diagnosing etiology

Principles of pain assessment

15

- Pain is highly subjective- must use **patient-oriented approach**
 - Ask & believe the patient (can use assessment tools)
- Comprehensive patient history & physical examination must be obtained
- **Mental factors** that alter pain threshold must be given attention
- Differentiating the type of pain
- **Assess & reassess pain intensity, relief & medications regularly**

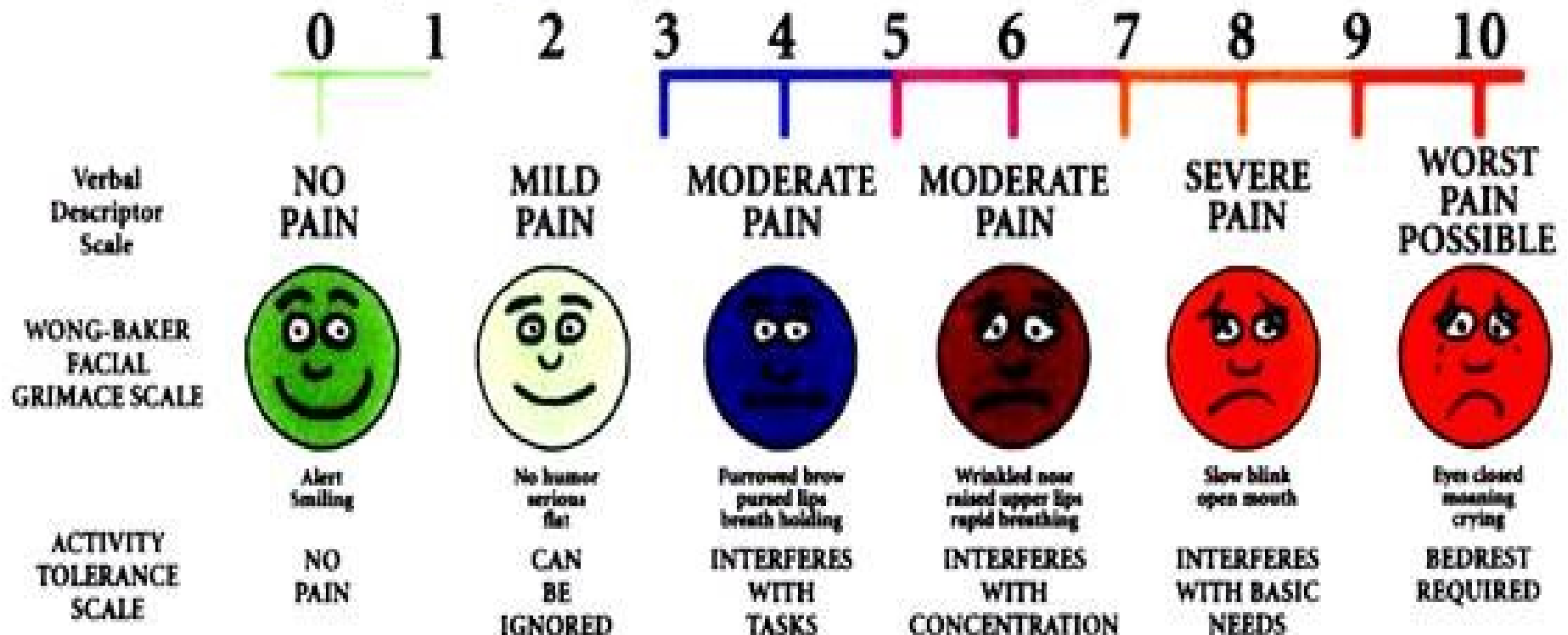
Assessment Tools

16

- Pain rating scales are used to evaluate **pain intensity** at baseline & follow up
 - Numeric rating scale(0-10 scale)
0=No pain **10**=Worst pain
 - **Pediatrics**-face of pain scale

UNIVERSAL PAIN ASSESSMENT TOOL

This pain assessment tool is intended to help patient care providers assess pain according to individual patient needs. Explain and use 0-10 Scale for patient self-assessment. Use the faces or behavioral observations to interpret expressed pain when patient cannot communicate his/her pain intensity.



Patients Pain History Assessment

PQRST characterization

18

P	Palliative factors	What makes the pain better?
	Provocative factors	What makes the pain worse?
Q	Quality	Describe the pain.
R	Radiation	Where is the pain?
S	Severity/intensity	How does this pain compare with other pain you have experienced?
T	Temporal factors	Does the intensity of the pain change with time?

Treatments

19

- Understand pain through the eyes of the patient
- Take time for a comprehensive pain assessment:
 - ▣ History and physical
 - ▣ Previous treatment
 - ▣ Psychosocial concerns
 - ▣ Past pain history
 - ▣ Comorbidities
 - ▣ Family history of addiction or pain challenges
 - ▣ Listen for common misconceptions about pain, treatment or fears
- Engage patient in developing pain treatment plan

Treatments...

20

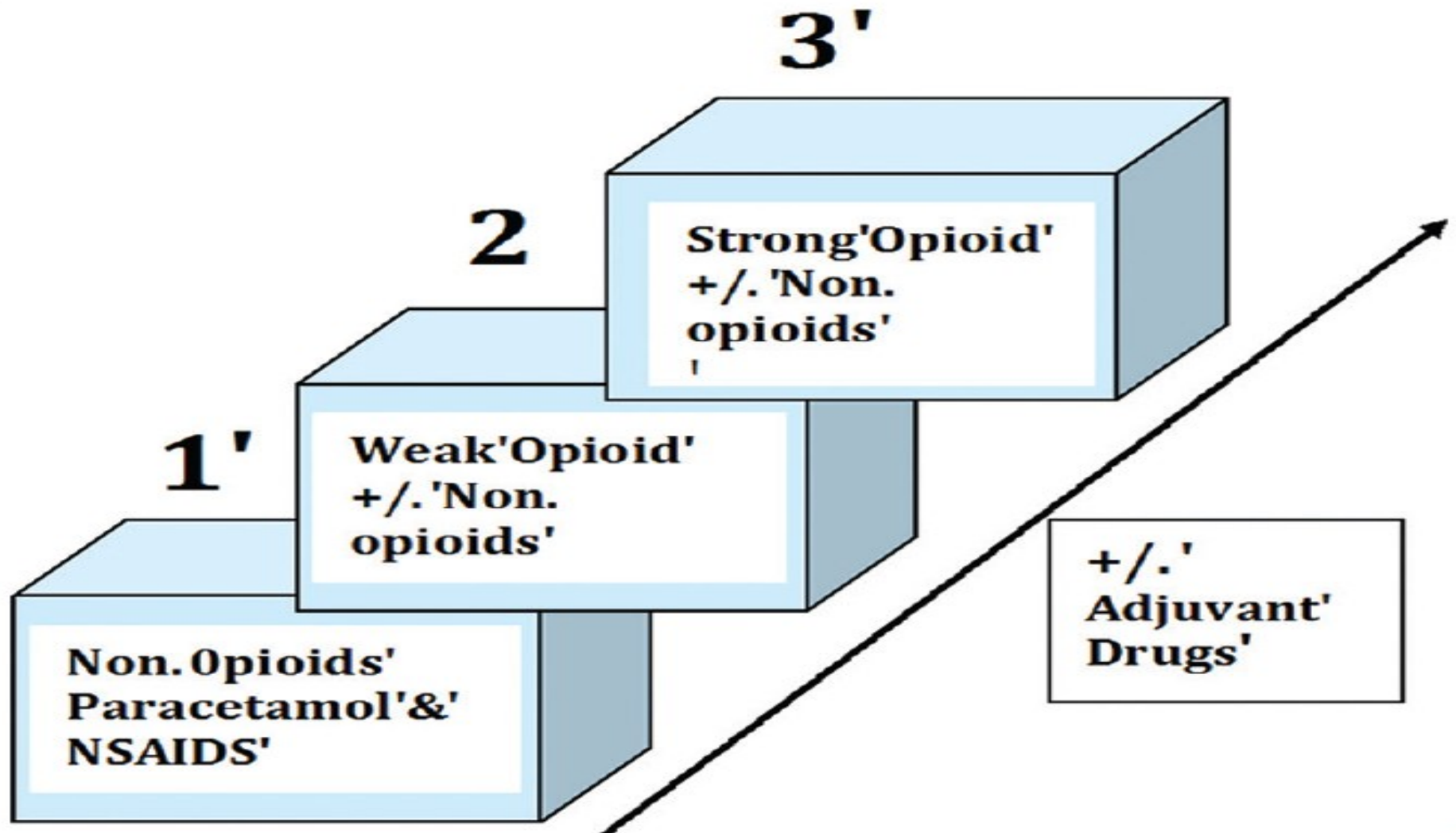
- **Total pain elimination is not a realistic goal,**
 - Instead, a more attainable objective is reducing pain to a tolerable predetermined level

Non pharmacological treatments

- Physical manipulation, application of heat or cold, massage, cognitive behavioural therapy, relaxation, acupuncture, and exercise are all modalities with variable efficacy for both acute and chronic pain

Pharmacological treatments(WHO LADDER)

21



Acute Pain Management

- Treatment is guided based on the severity of pain

Step 1: Mild pain (Rating scale 1-3)

Non opioid agents

- Paracetamol
 - NSAIDs
 - **Adjuvant:** antidepressants(TCA), anticonvulsants (Gabapentin), corticosteroid
- ± Adjuvant
- These agents especially **Gabapentin** and **TCA** are used for neuropathic pain

Cont...

23

Some notes about non-opioid agents:

- These drugs work by affecting prostaglandins production in response to pain stimulus
- The choice of one agent over the other depends on availability, cost, pharmacokinetics, pharmacological characteristics & side effect profile
 - Comparison data are inconsistent

Cautions

- NSAIDs in pts >70, renal insufficiency, heart failure & GI bleeding
- Paracetamol not to exceed 4 g-liver failure
- COX 2 inhibitors (celecoxib & rofecoxib) even if low risk of GI bleeding have greater risk for cardiac disorder.

Counseling Tips

25

Tip 1: Instruct patient to take NSAIDs with full glass of water and take after a meal

Tip 2: Patient should remain in an upright position for 15-30 min after administration

Tip 3: Avoid concurrent use of alcohol to minimize possible GI irritation

Cont...

26

Step 2: Moderate pain(4-7)

Weak opioid agents± Non opioid agents± Adjuvant

- **Weak opioid agents:** Codeine, hydrocodone, tramadol, dihydrocodeine, low dose oxycodone
- **Non-opioid:** aspirin, acetaminophen, NSAIDs
- **Adjuvant:** antidepressants, anticonvulsants, corticosteroids.....

Step 3: Severe pain (8-10)

Strong opioid agents± Non-opioid agents± Adjuvants

Strong opioid agents: Pethidine (meperidine), Morphine, fentanyl, hydromorphone, oxymorphone, methadone, high dose oxycodone

Facts about Opioids

28

- For moderate-severe acute pain, cancer related pain and chronic non-cancer pain
 - Generally opioids **don't completely eliminate pain but minimize its unpleasantness**
- All opioids share similar side effects even if the degree varies from agent to agent
 - **Constipation, sedation, Respiratory depression and N/V**
- PO is preferred but when immediate relief is required change to IV
- Treatment for acute pain should be around the clock rather than as per needed schedule.

Morphine

29

- Prototype opiate analgesic for moderate to severe pain
- Considered 1st line for severe pain
- **Two metabolites**
 - Morphin-6-glucuronide: analgesic
 - Morphine-3-glucuronide: responsible for side effect
- Available parenterally, orally or rectally
- First choice to treat pain associated with myocardial infraction because additionally it can dilate venous and arteriolar blood vessel

Con't

30

Meperidine or pethidine: 50-150 mg PO/IM/SC q3-4hr or PRN

- Not as potent as morphine
- Two problems
 - **Short half life**- needs larger dose for satisfactory pain relief
 - **In renal failure**-its metabolite **normeperidine** accumulates causing seizure
- Not usually recommended

Fentanyl

- Often used as an adjunct to general anesthesia
- More potent & shorter acting than meperidine
- Available as a **transdermal patch** for chronic pain providing pain relief for up to 72hrs.
 - Not useful for acute pain: Onset of analgesia 12-24hrs
- Also available as lozenge and buccal formulation for treatment of breakthrough cancer pain

Methadone

- Usually used to treat chronic pain including cancer pain but may also be used for acute pain
- Orally effective, extended duration of action & low cost
- May be proven to be effective in **neuropathic pain**

Major adverse effects of opioids

32

1. Constipation

- Decrease propulsive contraction of the GI tract resulting in reduced GI motility
- Need prophylaxis
 - Stimulant cathartics (e.g., senna or bisacodyl 5mg or 10 mg)
 - Stool softener may be concomitantly used
 - Occasionally, patients require enemas (process of liquid introduction into the rectum or colon)

2. Sedation

- Can be prevented or managed by
 - Reducing the dose of an opioid by 25-50% if analgesia is satisfactory
 - Lowering the requirement for opioids by the addition of a non-opioid analgesic or adjuvant analgesic,
 - Switching to another opioid,
 - Psychostimulants

3. Nausea & vomiting

- Through direct stimulation of CTZ
- More frequent **after initial dose & often subside with subsequent administration**
- Use standard antiemetic, such as promethazine, haloperidol, **metoclopramide**
- In some cases, ondansetron or meclizine can be helpful

Con't

35

4. Respiratory depression

- Usually at larger dose
- Reversed by the least amount of **naloxone** or using continuous infusion

5. Mental clouding

- **Low dose haloperidol or psychostimulants** for less agitated patients

6. Seizure

- Clonazepam or valproate
- Reduction in opioid dose
- Switch to a different opioid

7. Pruritus & Urinary retention

- Pruritus associated with histamine release
- Tolerance develop to the urinary retention over time

Opioids Switching

36

- **The need for switching:**
 - Pain controlled but intolerable side effects
 - Pain inadequately controlled but side effect prevent increase in analgesic dose
- When switching however must consider the comparable potency of the respective opioids

Dosing and Conversion Chart for Opioid Analgesics

37

<u>Drug Name</u> <u>Dose (mg)</u>	<u>Route</u>	<u>Equivalent</u>
▪ Morphine	IM	10
▪ Morphine 30	PO	
▪ Oxycodone	PO	30
▪ Hydromorphone	IM	1.5
▪ Hydromorphone	PO	7.5
▪ Meperidine	IM	75
▪ Meperidine	PO	300
▪ Methadone	IM	10
▪ Methadone	PO	20
▪ Fentanyl	IV	0.1
▪ Hydrocodone	PO	30

Principle of Treatment:

- Continual reassessment of pain and individualization of therapy
- The objective is to avoid fluctuation b/n severe pain & pain relief

Around the clock schedule: for maintenance
plus

As per needed schedule: for breakthrough pain

e.g. Morphine 15-30 mg q12h with 5-30 mg q4h as **PRN**

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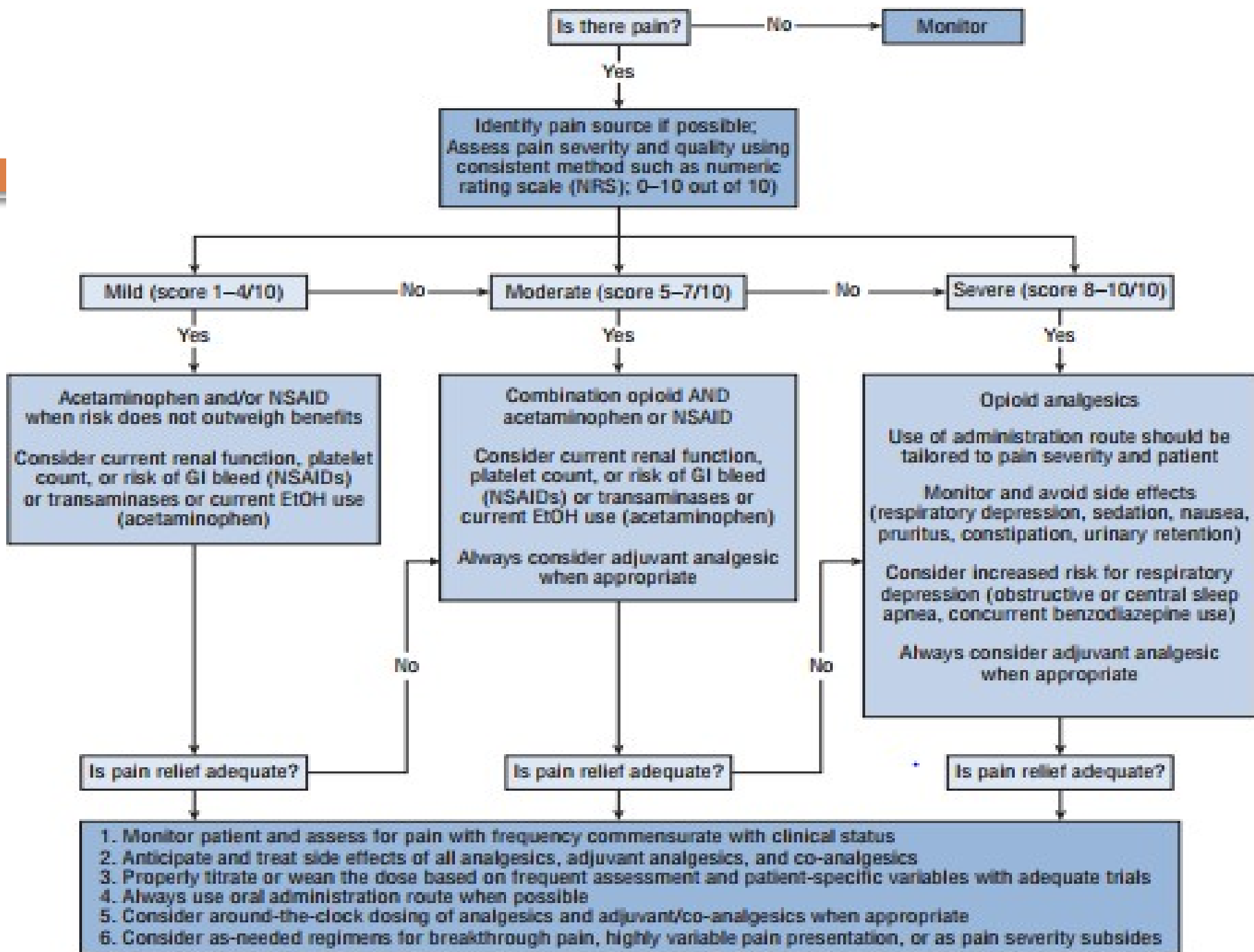
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- **Non-opioids are preferred as 1st line**
 - Especially **NSAIDs** for bone pain associated with metastasis
- The choice of opiates should be based on patients acceptance, efficacy, drug characteristics & side effect profile
- Use maximum dose before switching to the other
- Provide medications for the side effects of opioids
- Use adjuvants to maximize pain relief
 - **Neuropathic pain**-Anticonvulsants (gabapentin), & TCA
 - **Bone pain**- NSAIDs, Bisphosphonates

Anesthetics

40

- Regional analgesia with **local anesthetics** is useful for both acute and chronic pain.
- Commonly via injection (ie, in the joints, epidural or intrathecal space, nerve plexus, or along nerve roots) or applied topically.
- High plasma concentrations of local anesthetics can cause dizziness, tinnitus, drowsiness, disorientation, muscle twitching, seizures, and respiratory arrest.
- Cardiovascular effects include myocardial depression, heart block, and hypotension



Monitoring of Analgesics

42

Drug (class)	Adverse Reaction	Monitoring Parameter	Comments
Opioids	Respiratory depression	Respiratory rate OR end-tidal capnography	Capnography considered more sensitive; however, equipment may be expensive to incorporate. Higher risk—Obstructive sleep apnea, chronic obstructive pulmonary disease
	Constipation	Bowel movement frequency and consistency	Constipation may be assessed using Bristol Scale
	Sedation	Sedation scale	Will decrease over time
	Nausea, vomiting	Nausea, vomiting	Will decrease over time
	Tolerance	Regular efficacy monitoring	With chronic use, may lead to need for larger doses
	Dependence	Regular efficacy monitoring	Will develop with chronic use
	Addiction/abuse	Regular efficacy monitoring	Seldom problem with acute pain. In chronic pain use specified time trials with opioids, targeted functional end points, screening tools, and frequent monitoring
	Histamine release	Monitor for urticaria, pruritus, bronchospasm	Incidence varies among agents
	Increase in sphincter tone	Monitor for biliary spasm, urinary retention	Incidence varies among agents
	Hypogonadism	Monitor fatigue, depression, sexual dysfunction, amenorrhea (women)	Problem with chronic use
NSAIDs	Upper GI bleeding	Complete blood count, stool guaiac (if symptoms such as black tarry stools, warrant)	One of leading causes of hospitalizations due to drug-related adverse effects in the United States
	Acute renal failure	Serum creatinine	
Acetaminophen	Hepatotoxicity	Serum transaminases (ALT/AST) Liver synthesis tests (PT/INR, albumin) Acetaminophen serum concentration	Elevated transaminases may occur even at doses less than 4000 mg daily

THE END